Project title – Cataloguing a Library's Book Damage with Computer Vision

Data source – the 200 books in my library and the books in Randall Library. I will take pictures of the front covers and/or the spines of the books.

Existing projects done on the same data – None, I'm creating my own dataset.

Existing projects using similar techniques – Here's a blog post that creates a book inventory with images of the front covers of books, and here's a video about using AI to identify books from their spines. Here (paper1, paper2) are a few papers about using computer vision to detect civil infrastructure damage, which I could use to help me identify book damage, specifically the types of models they use and how they set them up. Here's an API that already identifies damage (to a very basic extent), but I want to build my own...

Team members – Me, myself, and I

Project goals –

1. Generate a dataset of books from my personal library and UNCW's Randall Library with labels to identify wear and general information about the book. Dataset will include both images of front covers (for an individual book) and spines (for multiple books)
2. Log the title, author, and the publisher and published date for each book, and explore categorization algorithms that identify the wear on the book (i.e. torn cover, loose spine).
3. Identify similar books to each book (either through categorizing the existing books logged [similar cover art / fonts] or prompting with GPT)

Potential challenges –

1. Lighting will probably be one of my greatest enemies. My library is very dark, but most public libraries are bright, so setting parameters so the algorithm can grab the information in both settings might be an issue.
2. Building a model (because I might need to make it from scratch) that can identify book wear. From what I can tell, there's not many algorithms already out there for this. If I had to make my own, I would have to include a broader dataset (because my books and the ones at the library aren't *that* damaged).
3. If I choose to use a model to find similar books, this will take some time. If I use a GPT prompt, there might be a few road bumps, but it'll probably be easier.
4. Getting all the information from the picture (finding a threshold that works well).
5. Setting up a database (potentially) to store the information for each book.